AAKASH SANGANI

Data Scientist · ML and LLM Systems

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SUMMARY

Data scientist, 3+ years across pharma analytics, fintech experimentation and AI governance. Builds and evaluates ML and LLM systems: edge inference, retrieval, Bayesian inference, fairness audits. Author of two open-source evaluation tools and an open agentic-failure benchmark.

TECHNICAL SKILLS

ML and LLM: RAG, agentic AI, LLM evaluation and red-teaming, MCP servers, Bayesian inverse RL, LSTM, XGBoost, SHAP, AIF360, scikit-learn, TensorFlow

Edge and inference: INT8 quantisation, ONNX, parallel batched inference, sentence-transformers, vector search, ASR / MT / TTS pipelines

Data and infrastructure: Python, SQL, pandas, NumPy, PostgreSQL, Apache Spark, A/B testing, AWS, Docker, Ansible, CI/CD gates, SARIF

Governance: EU AI Act, NIST AI RMF, ISO 42001, GDPR/DPDP, HIPAA, OWASP ASI, MITRE ATLAS

EXPERIENCE

AI Consultant Feb 2026 – present
Kalpa Impact Mumbai

- Quantised the speech and vision models to INT8 via ONNX and restructured inference into parallel batched streams on **Suno Sutra**, the open-source offline AI device from Digital India Bhashini (MeitY) and Current AI: peak memory down 40%, end-to-end latency ~8s → 2–3s across 22 Indic languages.
- Built and packaged **The Specialist**, a grounded offline retrieval module for that device: multilingual-e5-small embeddings with a deterministic TF-IDF fallback, two refusal gates, and a source citation on every answer.
- Designed the evaluation framework for the **VYOMA Innovation Challenge** (₹80 lakh, Bhashini × Current AI × Kalpa Impact): jury rubric, judging matrix and scoring thresholds. Verified edge-AI claims submitted to the Gates Foundation; wrote a section of *AGI Governance* with the Principal Scientific Adviser's Office on AI incident reporting.

Business Technology Solutions Associate → Senior Associate Sep 2023 – Jan 2026
ZS Associates Pune

- Launched an agentic AI analytics product for natural-language querying of sales-performance metrics.
- Owned incentive-compensation product logic for a Fortune 100 pharma client at 100% accuracy. Led a team of 4; top 1% performer firm-wide, promoted to Senior Associate within 8–10 months.

Product Analyst Intern Jul 2022 – Feb 2023
Bright Money Bangalore

- Built A/B testing infrastructure and SQL dashboards; launched a referral programme that saved $120K in customer-acquisition cost over 5 months.

OPEN-SOURCE PROJECTS

Inverse-Objective Regulator `github.com/aakash2410/Inverse-Objective-Regulator`

- Black-box auditor that infers a deployed agent's revealed objective from its tool-call traces via Bayesian inverse RL and scores divergence from the declared purpose, emitting JSON and SARIF mapped to the EU AI Act, OWASP, MITRE ATLAS and NIST AI RMF.
- Open 8-agent benchmark of agentic failure modes. A deterministic cross-encoder critic recovered planted divergence on 6 of 8 agents (mean r = 0.55) against a frontier LLM-as-judge (0.08 → 0.43), with near-zero variance and no API cost. Sole author of the arXiv system paper.

rag-compliance-auditor `github.com/aakash2410/MCP_RAG_Compliance`

- Sole-authored package on PyPI. MCP server certifying RAG pipelines against the EU AI Act, GDPR/DPDP, HIPAA, SEBI/RBI and NIST AI RMF across 7 dimensions; 8 MCP tools, RS256-signed certificates, runs as a CI gate with SARIF.

EDUCATION AND PUBLICATION

B.Tech, Information Technology, D. J. Sanghvi College of Engineering, University of Mumbai, 2019–2023. CGPA 8.90 / 10, full merit scholarship. Book chapter: Sangani, A., Doshi, P., & Sawant, V. "Unusual Social Media Behavior Detection Using Distributed Data Stream Mining," Auerbach (CRC Press), 2022–23.